

BELGAUM, India

WMO: 431980

Lat: 15.859N

Lon: 74.618E

Elev: 2487

StdP: 13.42

Time Zone: 5.50 (IND)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	55.5	57.7	34.3	31.8	85.0	37.6	36.2	83.5	18.3	79.3	16.8	79.2	0.3	90	0.285

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	25.0	97.7	66.9	95.9	67.1	94.2	67.2	75.7	85.3	74.8	83.7	74.1	82.4	6.6	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.4	136.2	79.0	72.6	132.7	78.1	72.0	129.7	77.2	41.2	85.9	40.3	84.2	39.6	82.6	84.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.2	16.1	13.9	DB	50.9	101.2	1.9	2.6	49.6	103.1	48.5	104.6	47.5	106.1	46.1	108.1	(4)
(5)			WB	46.4	79.0	1.9	2.0	45.0	80.5	43.9	81.6	42.8	82.8	41.4	84.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	76.2	71.8	75.7	80.3	83.2	82.3	76.7	74.1	73.9	74.8	75.8	73.9	71.9
	DBStd	4.43	2.68	2.97	3.08	2.45	2.69	3.04	1.91	1.79	2.09	1.99	2.59	2.66
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	9559	677	719	939	995	1001	802	746	741	744	800	716	679
	CDD65	4084	212	299	474	545	536	352	281	276	294	335	266	214
	CDH74	31730	1974	3149	5291	5835	5183	1844	716	751	1253	2117	1932	1685
	CDH80	13188	592	1469	2920	3263	2652	509	57	65	231	584	472	374
Wind	WSAvg	5.5	3.6	3.9	4.3	5.0	6.9	8.4	8.7	7.3	5.4	3.6	4.2	4.4
Precipitation	PrecAvg	40.7	0.1	0.1	0.5	1.7	3.5	6.2	11.0	7.3	4.6	4.1	1.4	0.2
	PrecMax	64.3	1.9	1.1	4.9	4.8	15.6	14.0	28.3	24.1	11.6	10.8	7.0	1.9
	PrecMin	22.0	0.0	0.0	0.0	0.0	0.2	0.8	2.5	2.2	0.4	0.1	0.0	0.0
	PrecStd	11.6	0.3	0.2	0.9	1.2	3.0	2.5	5.7	4.4	2.7	2.5	1.6	0.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	89.2	94.4	98.1	100.0	99.4	92.4	84.0	84.4	87.4	88.8	87.4	87.6
		MCWB	63.1	63.3	64.7	67.4	67.8	73.0	73.3	73.3	72.3	67.8	66.1	65.4
	2%	DB	87.1	92.2	96.2	98.2	96.9	88.2	81.6	81.8	84.7	86.5	85.5	85.2
		MCWB	62.9	63.2	64.6	67.0	68.9	73.3	72.6	72.8	72.4	68.3	66.4	64.4
	5%	DB	85.3	90.5	94.7	96.5	94.5	85.3	79.8	80.0	82.7	85.1	84.2	83.6
		MCWB	62.4	63.3	64.6	67.4	70.3	73.3	72.3	72.3	72.4	68.7	66.4	63.8
	10%	DB	83.7	88.5	93.0	94.7	92.2	82.6	77.9	78.1	80.7	83.6	83.0	82.1
		MCWB	62.2	63.2	64.9	68.0	71.2	72.9	72.1	72.0	72.1	69.4	66.4	63.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	70.0	71.1	74.3	76.5	77.2	76.3	74.6	74.6	75.6	75.1	73.2	71.5
		MCDB	79.9	83.4	84.2	87.2	88.4	84.7	80.4	81.2	82.8	82.6	80.8	81.5
	2%	WB	67.7	69.0	72.6	74.9	75.9	74.8	73.6	73.6	74.3	73.8	71.9	69.7
		MCDB	79.2	82.0	82.1	84.9	87.0	82.6	78.9	79.3	81.0	80.8	79.4	78.8
	5%	WB	66.1	67.5	71.2	73.8	74.9	74.1	72.9	72.9	73.4	73.0	71.0	68.4
		MCDB	77.6	81.0	80.9	83.3	85.7	81.5	77.6	78.0	79.7	79.6	78.3	77.3
	10%	WB	64.7	66.2	70.0	72.8	74.0	73.4	72.4	72.2	72.6	72.3	70.1	67.1
		MCDB	77.4	80.1	80.1	82.3	84.0	80.3	76.7	76.8	78.3	78.3	77.1	76.1
Mean Daily Temperature Range	5% DB	MDBR	24.4	26.4	26.6	25.0	21.7	11.6	8.0	8.8	12.4	16.0	19.4	22.3
		MCDBR	26.8	28.5	27.8	26.9	25.1	16.7	11.3	12.2	16.2	18.9	21.1	23.9
		MCWBR	8.9	9.3	9.5	8.9	7.2	4.9	4.3	4.9	5.8	6.3	7.4	8.5
	5% WB	MCDBR	22.1	25.6	25.5	24.6	21.6	13.3	8.9	10.0	12.9	14.6	15.5	18.6
	MCWBR	8.9	9.8	9.2	8.4	7.0	5.2	4.2	4.8	5.6	5.8	6.2	7.8	
Clear-Sky Solar Irradiance	taub	0.404	0.418	0.496	0.562	0.535	0.516	0.536	0.504	0.467	0.517	0.459	0.418	
	taud	2.239	2.224	2.016	1.890	1.981	2.078	2.011	2.112	2.232	2.025	2.148	2.218	
	Ebn at Noon	280	281	262	245	247	250	246	256	266	249	260	271	
	Edh at Noon	43	45	57	65	58	52	56	51	45	54	46	43	
All-Sky Solar Radiation	RadAvg	1674	1927	2088	2163	2060	1387	1168	1232	1447	1574	1576	1581	
	RadStd	67	59	82	75	114	126	127	83	137	86	99	57	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)	+0.27	N/A	+1.55	+0.67	+0.65	+0.61	N/A	N/A	N/A	N/A

Nomenclature: See separate page